

Project Definition Report

Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

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Summary

This is an example and template for an Project Definition Report (PDR) to be used when planning a Thesis.

[Text written like this contains generic notes and guidelines.]

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CHAPTER 1

Project Description

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, currently consisting of only around 1,500 documents - significantly smaller than that of its younger counterpart, which now spans more than 6,000 documents.¹ But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.² Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael), these are only available in print form.

This motivated the development of the palaeographical database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palaeographic analyses within the epigraphic corpus" (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known documents.^{3,4}

¹Steele 2023.

²Consiglio Nazionale delle Ricerche 2025.

³SigLA 2021a.

⁴SigLA 2021b.

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.⁵ According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation using modern computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

⁵Salgarella and Castellan 2020.

1.2 Prior work / Literature review

This thesis places itself in the intersection of computer vision, digital humanities, and palæography (the study of ancient writing). As it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. Consequently, the other two fields will merely provide the context and motivation for that research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM), 2. the palæographical and digital humanities work that this project will seek to complement (SigLA), and 3. the context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (Steele 2023), its new addition (Consiglio Nazionale delle Ricerche 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the palæographical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM are: its paper (Kirillov et al. 2023), its lighter implementation Mobile Sam (C. Zhang et al. 2023) - and the newest version of that one (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded deep learning historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018).

1.3 Research Question / Hypothesis / Problem statement

This section provides a very concise statement of your research focus. This can be formulated either as a ‘research question’, ‘hypothesis’, or ‘problem statement’. But you have to pick one or the other. The hypothesis or thesis is the highest-level problem or goal you are going to address. The specific list of problems — usually a handful, although sub-problems are sometimes given — is things that need to be solved if you are going to satisfy your hypothesis.

The context of the problem should already have been introduced in the prior sections. A general rule of thumb is that all keywords that are used in the problem statement or research question should have been introduced and explained in the text above.

Of course, the problem must be worthy of a thesis, and the importance of the problem should have been motivated in the text above. When the reader reaches this point, s/he should think that the stated problem indeed is one of the most important problems to solve in the context of the thesis. The problem should be stated unambiguously, and there is only *one* problem you solve (although there might be sub-problems to the overall problem).

In general, there are a few typical types of research you can do:

- You can try to *solve a real-world problem* for some real-world users or organizations.
- You can try to *solve a ‘theoretical’ problem* reported in the literature.
- You can try to *replicate* a scientific solution or method reported in the literature.
- You can develop a *novel approach or method* for solving a well-known problem

1.4 Research goals and methods

While the above section details the problems, your job is to translate those problems into research goals. Each goal should briefly indicate *how* you are going to solve the problem, i.e., the method you will use to solve it. Note that some authors sometimes combine problem statements, goals, and methods into a single section, while others separate them. Goals should be operational, i.e., if you later claim to achieve your goal, you should be able to match your solution against the goal statement.

I cannot overstate how important it is to have clear goals. Most examiners highlight these goals, and after reading the thesis, they then go back to see if you have accomplished your goals. If you have not, then you have a big problem with your thesis. Even worse are those where problems and goals are not clearly stated, for it means people are trying to evaluate your solutions in a vacuum.

1.5 Empirical considerations

Together, Ester Salgarella and I will compile a dataset of at least 25 images of original documents, sourced from the Cefael corpus, which will then be selected and cropped. For evaluation, their corresponding digitized layered drawings, stored as Krita (.kra) project files, will be used as ground truth. The ground truth data will then have to be cleared of any marked “deleted signs” or other annotation elements that our model is not intended to reproduce.

You should address questions like:

- Where do you get your data? Where do you analyze it? Can you do this yourself?
- How are you going to build some technology?
- How do you get access to users?
- How do you get access to the needed resources, such as money, hardware, software, lab space, ...?
- Who are you working with? A company, hospital, clinic, grassroots, ...
- What is your role? Are you expected to deliver something to an external partner?
- Are there any Intellectual Property Rights (IPR) issues involved?

The primary deliverable of this project will be a semi-automated annotation tool capable of generating initial segmentations from original Linear A document images and exporting researcher-friendly mask outputs for further refinement. All supporting components, including evaluation results, and analysis will be documented and presented in the written thesis.

1.6 Impact – innovation and application

The work of this thesis will present itself as a first proof-of-concept for the use of computer vision to accelerate the expansion of the SigLA database and the further digitization of the Linear A corpus. Its main contribution will be a semi-automatic annotation tool, which might prove useful to palæographers for generating initial segmentations of original document images, which can then be completed and refined more quickly while retaining accuracy. Hopefully, it will inspire further research and development in the field of digital epigraphy, particularly in the application of computer vision techniques to Linear A, and perhaps even in the broader context of small corpus ancient writing systems.

CHAPTER 2

Intended Learning Objectives

[This section should list your intended learning objectives (ILO) for this Thesis. A list of ILO is listed at <http://sdb.dtu.dk/2017/30/576> and included below. You should include and tailor most/some of their more generic ILOs, but also add some specific ones which are relevant for you and your project. Examples of specific ILO include learning a specific method or technology. You can be specific by adding a text starting with; “Specifically, the objective is to ...”.]

The intended learning objectives (ILO) of this Thesis are:

- can identify and reflect on technical scientific issues and understand the interaction between the various components that make up an issue
- can, on the basis of a clear academic profile, apply elements of current research at an international level to develop ideas and solve problems
- masters technical scientific methodologies, theories and tools, and has the capacity to take a holistic view of and delimit a complex, open issue, see it in a broader academic and societal perspective, and, on this basis, propose a variety of possible actions
- can, via analysis and modelling, develop relevant models, systems, and processes for solving technological problems
- can communicate and mediate research-based knowledge both orally and in writing
- is familiar with and can seek out leading international research within his/her specialist area.
- can work independently and reflect on own learning, academic development, and specialization

- masters technical problem-solving at a high level through project work, and has the capacity to work with and manage all phases of a project, including preparation of timetables, design, solution, and documentation

CHAPTER 3

Plan

3.1 Gantt Chart

Figure 3.1 shows an outline of the project, including the studying, creation, and evaluation of the pipeline in question.

3.2 Activities

[This section provides a table of the main phases and activities of the plan with a little description of each.]

Table 3.1: Project phases and activities.

#	Title	Description	Phase	Period
15	Project Definition	Writing up the PDR and planning the project. Also includes attending a start-up supervisor meeting.	Project Definition	W1–W2
23	Writing up Thesis	This is the final writing up – all of the chapters should be ready by now	Thesis	W26

3.3 Milestones

[This section provides a table of the main milestones of the plan with a little description of each. Examples of milestones include: (i) Approvals (e.g., of a UX design, data management, or of a study involving patients); (ii) Important deadlines (e.g., prototype ready for UX testing, recruitment of all patients, or final analysis of data done); (iii) Acquisition of core resources (e.g., getting access to hardware, setting up a lab experiment, or getting access to test data); and (iv) Thesis milestones (e.g., the PDR, chapters, and final Thesis).]

3.4 Deliverable

[This section provides a table of the main deliverables of the plan with a little description of each. Examples of deliverables include: (i) Design products (e.g., UX design,

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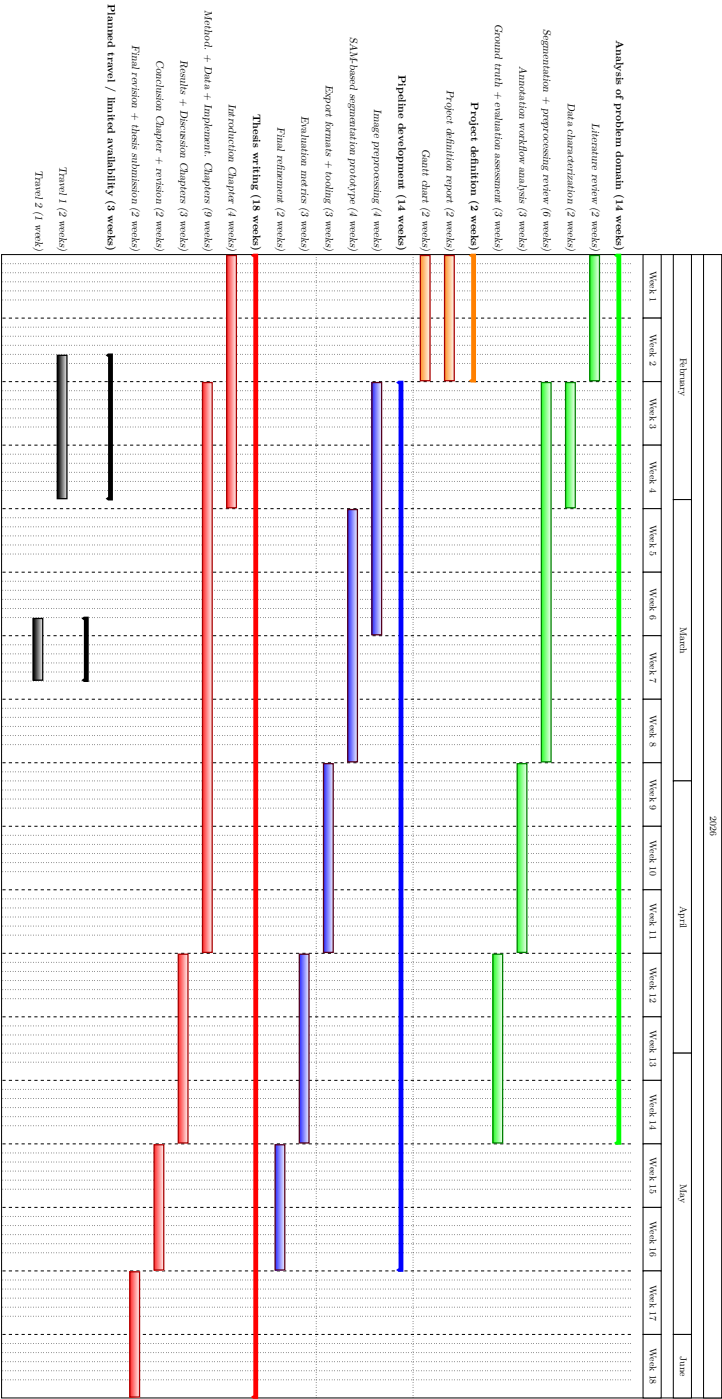


Figure 3.1: Project timeline (Gantt chart).

Table 3.2: Milestones. A: Approval, D: Deadline, AC: Acquisition, TH: Thesis.

#	Title	Description	Type	Week
M_1	Project Definition Report	Approval of the PDR.	TH	4
M_2	Milestone 2	This is the second milestone....	D	5
M_x	Milestone x	This the x'th milestone....	D	N
F	Thesis hand-in	Hand-in of final Thesis	TH	T

a requirement analysis, a design model, a data model), (ii) Hardware and Software (e.g., a UX prototype, a lab setup, an app for download in App Store); (iii) Data analysis (e.g., machine learning analysis or statistical analysis of user study data); and (iv) Thesis deliverables (e.g., chapters, figures).]

Table 3.3: Deliverable. R: Report, SW: Software, ST :Study.

#	Title	Description	Type	Week
D_1	Project Definition Report	Final version of the PDR.	R	4
D_2	Milestone 2	This is a software milestone....	SW	8
D_2	Final Thesis	Final proof-read Thesis ready to hand in	R	T

3.5 Risk Analysis

[A risk analysis identifies the top-5 or so risks that can significantly upset the entire project. Typical risk factors are often those that are not under your own control. Examples include getting access to data from someone else, the ability to run some experiments that will generate data, getting access to hardware/software that is needed, and getting access to people, such as company representatives, users, and patients.]

Table 3.4: Risks. 1:low risk, 5:high risk of delays.

#	Title	Description	Risk	Week
R_1	Risk no. 1	This is a high risk. Describe here how to mitigate it.	5	4
R_2	Risk no. 2	This is a low risk. Describe here how to mitigate it.	2	4

APPENDIX **A**

Acronyms

SAM	Segment Anything Model
Cefael	Collections de l'École française d'Athènes en ligne
PDR	Project Definition Report
ILO	intended learning objectives
IPR	Intellectual Property Rights

APPENDIX B

An Appendix

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